PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Treatment Outcomes of Blastocysts Thaw-Cycles, Comparing the Presence and Absence of a Corpus Luteum: A Systematic Review and Meta-analysis.
AUTHORS	Gan, Joscelyn; Rozen, Genia; Polyakov, Alex

VERSION 1 – REVIEW

REVIEWER	Hart, Roger School of Womens and Infants Health, University of Western Australia
REVIEW RETURNED	21-Jul-2021

GENERAL COMMENTS	Interesting meta-analysis only one very minor comment that should be included with more weighting (as it is mentioned) is that with the retrospective studies the choice of use a HRT approach rather than a modified natural cycle may well be heavily influenced by the reason for subfertility eg an older woman, or a woman with PCOS may be more likely to be directed to the HRT approach- both these cohorts have greater risks of miscarriage. This is acknowledged but this should be stated more prominently or at least discussed to a greater degree.

REVIEWER	Anderson, Richard University of Edinburgh
REVIEW RETURNED	26-Jul-2021

GENERAL COMMENTS	This paper presents a meta-analysis of pregnancy outcomes following frozen embryo transfer in either natural cycles or artificial cycles. The authors highlight that natural cycles are associated with the presence of a corpus luteum and therefore use this distinction in their title and analysis. While ovulatory cycles of course imply the presence of a corpus luteum, this distinction from the way the literature has been described previously is novel but I am not sure that it completely captures the difference between natural cycles and artificial cycles. The presence of a corpus luteum would be an important part of a discussion, but I am not sure it merits the prominence the authors have given it here. The other novelty of the paper is to focus only on blastocyst transfers, which is valuable and aligns with current practice. Of the nine studies included, only two were RCTs with the rest being retrospective cohort studies and thus prone to bias. The authors highlight this in the discussion, particularly highlighting the importance of comparing normal ovulatory women with women with PCOS where other factors including BMI may have significant impact on pregnancy outcomes, eg miscarriage rates.

	The methodology is carefully described and bearing in mind the limitations of the underlying studies is fully described. A key outcome is Beta hcg rates, laboratories in fact measure hcg not Beta hcg, please correct this throughout. On page 11, it would be helpful to point out that the two RCTs were both very small and specify in the text the number of patients included from these two trials versus the much larger retrospective studies. The paragraph on clinical pregnancy rates at line 29 includes a paragraph of discussion which would be best removed from the results section. I think the authors are also overinterpreting their data by thinking that the true likely point estimate lies between the point estimates of their main analysis and the sensitivity analysis which involved removing the one study that pointed to a different conclusion.

REVIEWER	Phan, Thuan University of Liverpool, Department of Women's and Children's Health
REVIEW RETURNED	14-Nov-2021

GENERAL COMMENTS	The specific goal of this paper is to mainly look at the outcomes of thaw cycles, drawing a comparison between the presence and the absence of corpus luteum. It appears to be the first review aiming to the treatment outcome. The authors should be congratulated for conducting an important and well-designed systematic review and meta-analysis with obvious statistical analysis alongside reasonable interpretation. The manuscript is well written and carefully follows the PRISMA checklist which is highly recommended for reporting a systematic review. Regarding statistical assessment, the manuscript should be considered the high-quality scientific one that includes accurate and rigorous statistical and data analyse alongside obvious description and explanation of chosen statistical methods. The authors have also presented enough information to verify their calculation and there is not any results interpreted beyond the range of data. To sum up, I highly recommend that the content of this manuscript will have value for publication. Minor points requiring revision: - Was there any language restrictions in the search? It causes confusion due to minor conflict between line 3-5 / page 8 and line 46-51/ page 8. - Line 35-42/page 9, there is a duplicate sentence. - Yellow symbol (!) requires definition, page 38, line 55-57 - Supplementary Table IV seems not to be very important for the manuscript. In my opinion, it should be withdrawn.

VERSION 1 – AUTHOR RESPONSE

Reviewer #1: Interesting meta-analysis	Thank you for your time in carefully reviewing	Page 21, line 8-15: However, cycles without a CL are still important as they may be necessary for women

only one very minor comment that should be included with more weighting (as it is mentioned) is that with the retrospective studies the choice of use a HRT approach rather than a modified natural cycle may well be heavily influenced by the reason for subfertility eg an older woman, or a woman with PCOS may be more likely to be directed to the HRT approach- both these cohorts have greater risks of miscarriage. This is acknowledged but this should be stated more prominently or at least discussed to a greater degree.	our manuscript. We agree with your suggestion and the importance of highlighting this limitation.	with irregular or absent periods and for cycles involving donor oocytes. As a result of this and the retrospective study design of many of the included studies, it should be noted that the population in whom artificial thaw cycles are performed may have an inherently different, possibly higher, risk of pregnancy losses. However, the AC approach is used routinely in many centres and therefore would not be subject to this bias.
Reviewer 2	**Response**	**Changes in text**
This paper presents a meta-analysis of pregnancy outcomes following frozen embryo transfer in either natural cycles or artificial cycles. The authors highlight that natural cycles are associated with the presence of a corpus luteum and therefore use this distinction in their title	Thank you very much for taking the time to review our manuscript and share your expertise with us. We decided to take the approach we did after carefully reviewing the available literature. We noted that an emerging area of research was the role of the corpus luteum – beyond the	

and analysis. While ovulatory cycles of course imply the presence of a corpus luteum, this distinction from the way the literature has been described previously is novel but I am not sure that it completely captures the difference between natural cycles and artificial cycles. The presence of a corpus luteum would be an important part of a discussion, but I am not sure it merits the prominence the authors have given it here.	functions and roles traditionally held, i.e. progesterone production. After much deliberation, we decided it would be a novel and interesting approach to compare cycles with and out a CL to see if there were any notable trends.	
The other novelty of the paper is to focus only on blastocyst transfers, which is valuable and aligns with current practice.		

Of the nine studies included, only two were RCTs with the rest being retrospective cohort studies and thus prone to bias. The authors highlight this in the discussion, particularly highlighting the importance of comparing normal ovulatory women with women with PCOS where other factors including BMI may have significant impact		

on pregnancy outcomes, eg miscarriage rates.		
The methodology is carefully described and bearing in mind the limitations of the underlying studies is fully described.		
A key outcome is Beta hcg rates, laboratories in fact measure hcg not Beta hcg, please correct this throughout.	Thank you for this suggestion, we have made the appropriate changes throughout the manuscript.	Please see the follow pages and line numbers which correspond with all changes: Page 5, line 28 Page 9, multiple lines Page 10, multiple lines Page 14, multiple lines Page 14, line 258-259 Page 16, line 60 Page 18, line 12 Table 1 Figure 2 (change title) Supplementary file 6 & 8
On page 11, it would be helpful to point out that the two RCTs were both very small and specify in the text the number of patients included from these two trials versus the much larger retrospective studies.		Page 10, line 42-47: Two of which were randomised controlled trials (RCTs)(22, 23) both of which studied small sample sizes. The remaining seven were retrospective cohort studies(10, 16, 20, 21, 24-26) which followed a much larger sample size.
The paragraph on clinical pregnancy rates at line		Page 16, Line 20-28; line: Based on these two

29 includes a paragraph of discussion which would be best removed from the results section. I think the authors are also overinterpreting their data by thinking that the true likely point estimate lies between the point estimates of their main analysis and the sensitivity analysis		analyses, it can be inferred that the most likely point estimate lays somewhere between 1.06 and 1.12, favouring cycles with CL. The confidence interval of this point estimate may include 1, but there is a trend towards cycles with CL resulting in a higher clinical pregnancy rate. While statistical significance may not be demonstrable, this finding may be likely to be

which involved removing the one study that pointed to a different conclusion.		clinically significant. Subgroup analysis of clinical pregnancy rates by study design is shown in Supplementary Figure 2.
Reviewer 3	**Response**	**Changes in text**
The specific goal of this paper is to mainly look at the outcomes of thaw cycles, drawing a comparison between the presence and the absence of corpus luteum. It appears to be the first review aiming to the treatment outcome. The authors should be congratulated for conducting an important and well-designed	Thank you for reviewing our manuscript and suggesting valuable feedback to improve it further.	

systematic review and meta-analysis with obvious statistical analysis alongside reasonable interpretation. The manuscript is well written and carefully follows the PRISMA checklist which is highly recommended for reporting a systematic review. Regarding statistical assessment, the manuscript should be considered the high-quality scientific one that includes accurate and rigorous statistical and data analyse alongside obvious description and explanation of chosen statistical methods. The authors have also presented enough information to verify their calculation and there is not any results interpreted beyond the range of data. To sum up, I highly recommend that the content of this manuscript will have value for publication.		
Was there any language restrictions in the search? It causes confusion due to minor conflict between line 3-5 / page 8 and line 46-51/ page 8.	There were no language restrictions in the search. In the search we screened through papers available in other languages, most of which had titles and abstracts available in	Page 9, line 6-9: removed.

	English. However, even after reviewing these titles/abstracts, we found that they were not	

	relevant to our question hence they were excluded. We have revised the exclusion criteria to reduce conflict and increase clarity.	
Line 35-42/page 9, there is a duplicate sentence.	Thank you for bringing this to our attention. We have removed the duplicate sentence.	Page 9, line 43-46 sentence deleted: Where biochemical pregnancy was not reported, it was calculated by subtracting the reported clinical pregnancies from the number of positive hCG results.
Yellow symbol (!) requires definition, page 38, line 55-57	The yellow symbol (!) is the overall score of the studies, as it is in yellow there are some concerns as to the potential risk of bias in the study designs of the two included RCTs. We agree the legend should be updated to better communicate this.	Supplementary file 2: Updated legend.
Supplementary Table IV seems not to be very important for the manuscript. In my opinion, it should be withdrawn.	We are happy to withdraw the table, but we can make it available if the need arrives.	Supplementary Table IV removed.

VERSION 2 – REVIEW

REVIEWER	Anderson, Richard University of Edinburgh
REVIEW RETURNED	21-Dec-2021

GENERAL COMMENTS	Thank you for these revisions, I have no further comments.